

Investigating Use of Stirling's Formula In Asymptotic Approximation Of Cosine

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1 Definitions

The Gamma Function, $\Gamma(x)$, is an important generalization of factorials and is firstly defined as:

$$\Gamma(x) = \int_0^{\infty} t^{x-1} e^{-t} dt$$

Where recursive iterations of integration-by-parts lends the property $\Gamma(x+1) = x\Gamma(x)$, in which parallels the $(n+1)! = (n+1)n!$ principal defining the integer factorials. This, along with $\Gamma(1)$ gives rise to the definition $\Gamma(x) = (x-1)!$.

However, the Gamma function moreover possesses a number of additional definitions and representations, one of which, the Euler Product Definition, will be utilized here.

$$\Gamma(x) = \frac{1}{x} \prod_{k=1}^{\infty} \frac{\left(1 + \frac{1}{k}\right)^x}{1 + \frac{x}{k}}$$

From this product, and with knowledge of the infinite product definition of sine as;

$$\sin(\pi x) = \pi x \prod_{k=1}^{\infty} \left(1 - \frac{x^2}{k^2}\right)$$

One can verify the famed *Reflection Formula* for the Gamma function via combining products. Specifically,

$$\begin{aligned} \Gamma(x)\Gamma(1-x) &= \Gamma(x)(-x)\Gamma(-x) = (-x) \left(\frac{1}{x} \prod_{k=1}^{\infty} \frac{\left(1 + \frac{1}{k}\right)^x}{1 + \frac{x}{k}} \right) \left(\frac{1}{-x} \prod_{k=1}^{\infty} \frac{\left(1 + \frac{1}{k}\right)^{-x}}{1 + \frac{-x}{k}} \right) \\ \Gamma(x)\Gamma(1-x) &= \frac{1}{x} \prod_{k=1}^{\infty} \frac{1}{1 + \frac{x}{k}} \frac{1}{1 + \frac{-x}{k}} = \frac{1}{x} \prod_{k=1}^{\infty} \frac{1}{1 - \frac{x^2}{k^2}} = \frac{1}{\frac{\sin(\pi x)}{\pi}} = \frac{\pi}{\sin(\pi x)} \end{aligned}$$

With slight rearrangement, this will be the starting point of our examination: (Cosine will be used from here further for a simpler algebraic process, though note sine will shortly reappear in the result.)

$$\begin{aligned} \frac{\pi}{\Gamma(x)\Gamma(1-x)} &= \sin(\pi x) \\ \sin\left(\pi\left(x + \frac{1}{2}\right)\right) &= \frac{\pi}{\Gamma\left(x + \frac{1}{2}\right)\Gamma\left(1 - x - \frac{1}{2}\right)} = \cos(\pi x) \end{aligned}$$

Next, let us introduce Stirling's Approximation of the factorial as $x! \sim \sqrt{2\pi x} \left(\frac{x}{e}\right)^x$, where equivalence to the factorial is achieved to an increasingly greater degree as x tends to positive infinity, and who's approximation is quite precise even outside of the asymptotic scale.

One approach to a derivation is to systematically approximate the Gamma function to simplify the integration. First,

$$x! = \int_0^\infty t^x e^{-t} dt = \int_0^\infty e^{x \ln(t)} e^{-t} dt$$

Substituting $t = xu, dt = xdu$;

$$xe^{x \ln(x)} \int_0^\infty e^{x(\ln(u)-u)} du$$

And by implementing the Taylor series of $\ln(x) - x$ at a local maximum of $x = 1$ as $-1 - \frac{(x-1)^2}{2}$, one attains an integrand roughly equal to the Gaussian Integral. The rapidly decreasing $e^{-\frac{x}{2}(u-1)^2}$ term has a negligible difference to the y-axis centered Gaussian, but as the x coefficient gains size, this difference quickly vanishes.

$$xx^x e^{-x} \int_0^\infty e^{-\frac{x}{2}(u-1)^2} du \sim xx^x e^{-x} \cdot \sqrt{\frac{\pi}{\frac{x}{2}}} = \left(\frac{x}{e}\right)^x \sqrt{2\pi x}$$

2 Manipulation

Adjoining the Reflection Formula, the $\Gamma(x) = (x-1)!$ property, as well as Stirling's approximation one obtains:

$$\begin{aligned} \cos(\pi x) &= \frac{\pi}{(x - \frac{1}{2})! (-x - \frac{1}{2})!} \sim \frac{\pi}{\sqrt{2\pi (x - \frac{1}{2})} \left(\frac{x - \frac{1}{2}}{e}\right)^{(x - \frac{1}{2})} \sqrt{2\pi (-x - \frac{1}{2})} \left(\frac{-x - \frac{1}{2}}{e}\right)^{(-x - \frac{1}{2})}} \\ &= \frac{1}{2\sqrt{\frac{1}{4} - x^2} \cdot e^{(-x - \frac{1}{2})} (x^2 - \frac{1}{4})^{\frac{(-2x-1)}{2}} (x - \frac{1}{2})^{2x}} \\ &= \frac{1}{2e \left(\frac{1}{4} - x^2\right)^{-x} (x - \frac{1}{2})^{2x}} \\ C(x) &= \frac{\left(\frac{1+2x}{1-2x}\right)^x}{2e} \end{aligned}$$

2.1 Plots

Due to the base of this exponentiation being negative for all x outside of $-0.5 \leq x < 0.5$, the $C(x)$ function seems to map *most* real axis inputs to complex values. Ergo, plotting the real and imaginary components against the x-axis can be used to visualize our cosine approximation, as depicted within the figure below.

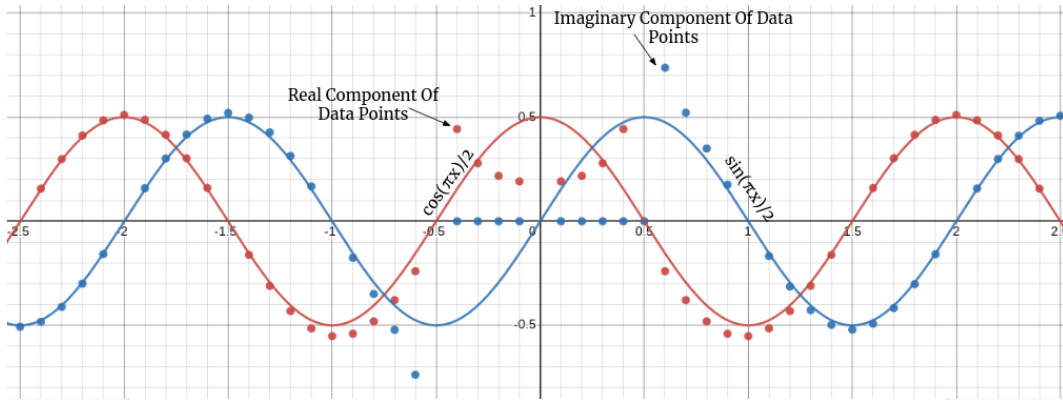


Figure 1: Real And Imaginary Parts Of C(x)

Indeed, the adherence of $C(x)$ to both of the displayed trigonometric functions *improves* with greater x value inputs (a feature absent of a Taylor expansion, intriguingly), as would be expected of the Stirling's approximation which generated it. Interestingly however, the height of both cosine and sine trend-lines are precisely half of the magnitude of the original cosine which produced $C(x)$.

3 Evaluating The Limit

Conjecturing from the findings of Section 2.1, one may wish to directly relate the three relevant functions as components of the following approximation: (in which the Euler Identity, $e^{i\theta} = \cos(\theta) + i \sin(\theta)$, is applied)

$$\frac{\left(\frac{1+2x}{1-2x}\right)^x}{2e} \sim \frac{\cos(\pi x)}{2} + \frac{i \sin(\pi x)}{2} = \frac{1}{2}e^{i\pi x}$$

If so, then;

$$\lim_{x \rightarrow \infty} \left(\frac{1+2x}{1-2x}\right)^x = e^{i\pi x + 1}$$

$$\lim_{x \rightarrow \infty} \left(\frac{1+2x}{1-2x}\right)^x = e^{\lim_{x \rightarrow \infty} \ln\left(\left(\frac{1+2x}{1-2x}\right)^x\right)} = e^{\lim_{x \rightarrow \infty} \ln((-1)^x \left(\frac{1+2x}{2x-1}\right)^x)}$$

$$= e^{\lim_{x \rightarrow \infty} [x \ln(-1) + \ln\left(\left(1 + \frac{2}{2x-1}\right)^x\right)]} = e^{\lim_{x \rightarrow \infty} \left[x \ln(-1) + \ln\left(\left(1 + \frac{1}{x-\frac{1}{2}}\right)^x\right)\right]} = e^{\lim_{x \rightarrow \infty} [x \ln(-1) + \ln(e)]} = e^{\lim_{x \rightarrow \infty} [xi\pi + 1]}$$

Where $\ln(-1)$ is computed as $\ln(e^{i\pi}) = i\pi$ by the Euler Identity, and where the exponential definition $\lim_{x \rightarrow \infty} \left(1 + \frac{z}{x}\right)^x = e^z$ is substituted into the calculation. Despite the fact that the remaining limit is of course indeterminate, since $e^{i(\pi x + -i)}$ merely equates to a $\cos(\infty - i) + i \sin(\infty - i)$ expression, it's theoretical existence closely aligns to the expected exponential.

By an identical methodology, the generalized form $\lim_{x \rightarrow \infty} \left(\frac{1+kx}{1-kx}\right)^x$ may be computed as:

$$\lim_{x \rightarrow \infty} \left(\frac{1+kx}{1-kx}\right)^x = e^{\lim_{x \rightarrow \infty} \ln((-1)^x \left(\frac{1+kx}{kx-1}\right)^x)} = e^{\lim_{x \rightarrow \infty} \ln((-1)^x \left(1 + \frac{2}{kx-1}\right)^x)} = e^{xi\pi + \ln\left(e^{\frac{2}{k}}\right)} = e^{(xi\pi + \frac{2}{k})}$$

From which gives rise to the approximation $e^{-\frac{2}{k}} \left(\frac{1+kx}{1-kx}\right)^x \sim e^{xi\pi}$ or

$$\left(\frac{1+kx}{1-kx}\right)^x \sim e^{\frac{2}{k}} (\cos(\pi x) + i \sin(\pi x))$$

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